Predict low review using delivery features

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Packages

Purpose

We trained a ridge-regularized logistic regression (L2 penalty; glmnet, alpha = 0) to predict low review (is_low_review = 1) using delivery delay buckets, price, freight, item count, and month. Ridge regularization was used to ensure stable estimation under potential (quasi-)separation and to improve generalization.

```r
library(DBI)
library(RPostgres)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(glmnet)
```

```
## Loading required package: Matrix

## Loaded glmnet 4.1-10
```

```r
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.

##
## Attaching package: 'pROC'

## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

Load data from Postgres

Quick sanity checks

```
## [1] 95830     7
```

```
##                           order_id is_low_review delay_bucket log_price
## 1 00010242fe8c5a6d1ba2dd792cb16214             0            0  4.092677
## 2 00018f77f2f0320c557190d7a144bdd3             0            0  5.484382
## 3 000229ec398224ef6ca0657da4fc703e             0            0  5.298317
## 4 00024acbcdf0a6daa1e931b038114c75             0            0  2.638343
## 5 00042b26cf59d7ce69dfabb4e55b4fd9             0            0  5.302807
## 6 00048cc3ae777c65dbb7d2a0634bc1ea             0            0  3.131137
##   log_freight_value items_count month
## 1          2.659560           1     9
## 2          3.041184           1     4
## 3          2.937573           1     1
## 4          2.623944           1     8
## 5          2.951780           1     2
## 6          2.616666           1     5
```

Data cleaning + feature engineering

Goal: - Keep complete cases for modeling - Cap extreme delays (reduce outlier impact) - Log-transform skewed monetary variables - Bucket delay to make the relationship more stable and reduce separation risk

Train / test split

Design matrix for glmnet

glmnet expects an X matrix (numeric) and y vector (0/1)

Ridge logistic with k-fold CV to pick lambda

Why ridge (L2)? - Logistic regression may fail to converge under (quasi-)separation - Ridge shrinks coefficients, stabilizes estimation, and improves generalization

```
## [1] 0.008269392
```

```
## [1] 0.05315625
```

Evaluate on test set using AUC

AUC = 0.704 on the test set, indicating the model has meaningful ability to rank low-review orders above non-low-review orders (above random baseline of 0.5).

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
## Area under the curve: 0.707
```

```
## Area under the curve: 0.7044
```

Choose a decision threshold (turn prob -> 0/1)

Threshold controls the precision/recall trade-off. - Higher threshold: fewer false positives, but more false negatives - Lower threshold: more true positives, but more false positives Decision threshold and confusion matrix

Because the dataset is imbalanced (low reviews are ~12.8% of the test set), we evaluated classification performance at a practical threshold. We selected threshold = 0.20 to balance precision and recall (more balanced than 0.50, and far fewer false positives than the Youden threshold).

At threshold = 0.20, the test confusion matrix was:

TP = 809, FP = 460, FN = 1643, TN = 16254 (N = 19,166)

Derived metrics: Precision = 0.638 Recall (Sensitivity) = 0.330 Specificity = 0.972 Accuracy = 0.890 F1-score = 0.435

Interpretation: With a 0.20 threshold, the model identifies about one-third of low-review cases (recall ~33%) while keeping false alarms relatively limited (precision ~64%, specificity ~97%). This threshold provides a more balanced operating point than the default 0.50 in an imbalanced setting.

```
##        y_test
## pred_05     0     1
##       0 16530  1806
##       1   184   646
```

```
##        y_test
## pred_03     0     1
##       0 16459  1739
##       1   255   713
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
##   threshold sensitivity specificity
## 1 0.1261443   0.4473899   0.8882374
```

```
##            y_test
## pred_youden     0     1
##           0 14846  1355
##           1  1868  1097
```

```
##        y_test
## pred_02     0     1
##       0 16254  1643
##       1   460   809
```

Show key terms first (delay buckets + month + numeric vars)

1.Low-review risk increases sharply once delivery is delayed beyond ~4 days. Delay buckets show a steep rise in odds from the 4–7 day range onward, indicating customer dissatisfaction escalates once delays move into multi-day territory.

2.Late delivery is the dominant predictor, with freight as a secondary contributor. Delay buckets are by far the strongest signals, while freight cost (log_freight) also shows a positive association with low-review risk—suggesting delays combined with higher shipping costs are key drivers of dissatisfaction.

3.Other factors have relatively minor impact. Month effects are small (seasonality is mild), price has a negligible effect, and items_count contributes little after regularization.

```
##                 term         beta  odds_ratio
## 1        (Intercept) -3.277477590  0.03772329
## 2     delay_bucket1-3  1.150719358  3.16046560
## 5   delay_bucket15-30  2.645488787 14.09033056
## 3     delay_bucket4-7  2.315311613 10.12807847
## 4    delay_bucket8-14  2.727458441 15.29396706
## 8         items_count  0.000000000  1.00000000
## 7   log_freight_value  0.299944649  1.34978409
## 6           log_price  0.043867730  1.04484414
## 17            month10 -0.050051369  0.95118056
## 18            month11  0.076813859  1.07984105
## 19            month12  0.092211822  1.09659708
## 9              month2  0.150830466  1.16279951
## 10             month3  0.133953636  1.14333981
## 11             month4 -0.001948809  0.99805309
## 12             month5 -0.094356220  0.90995856
## 13             month6 -0.077360283  0.92555633
## 14             month7 -0.134725491  0.87395579
## 15             month8 -0.169427696  0.84414779
## 16             month9 -0.121488778  0.88560099
```

```
##                 term       beta odds_ratio
## 4    delay_bucket8-14  2.7274584 15.2939671
## 5   delay_bucket15-30  2.6454888 14.0903306
## 3     delay_bucket4-7  2.3153116 10.1280785
## 2     delay_bucket1-3  1.1507194  3.1604656
## 7   log_freight_value  0.2999446  1.3497841
## 15             month8 -0.1694277  0.8441478
## 9              month2  0.1508305  1.1627995
## 14             month7 -0.1347255  0.8739558
## 10             month3  0.1339536  1.1433398
## 16             month9 -0.1214888  0.8856010
```

convert to probability

Using the training-set baseline (9.2% low-review probability for on-time deliveries), the model's odds ratios imply that low-review risk rises to ~24% for 1–3 days late and to ~51% once delays reach 4–7 days, exceeding ~59–61% for delays beyond 8 days (holding other factors constant).

```
##                        term odds_ratio       prob
```

```
## 1 delay_bucket0 (baseline)   1.000000 0.09222188
## 2           delay_bucket1-3   3.160466 0.24304021
## 3           delay_bucket4-7  10.128078 0.50712678
## 4          delay_bucket8-14  15.293967 0.60841530
## 5         delay_bucket15-30  14.090331 0.58872237
```